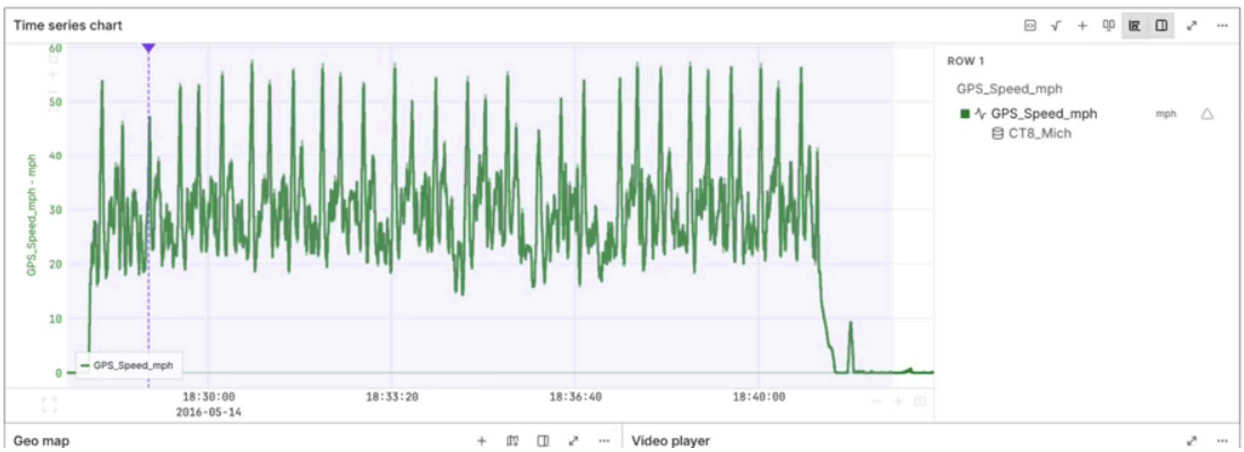
How to Transform a Time Series

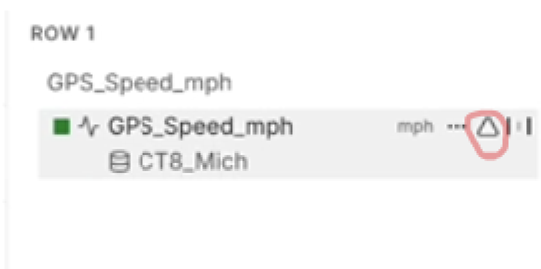
In this tutorial, we will go over how to **transform** a time series.**Transform** is an operation, such as absolute value or cumulative aggregate, applied to a variable in our time series. You can apply to an existing variable or create a new derived variable. **Transforms** can stack, and you can edit or remove them.

In this guided example, we will transform a racing channel with GPS speed data. We will transform the data to find the derivative of speed (acceleration) then stack another transform derivative (jerk).

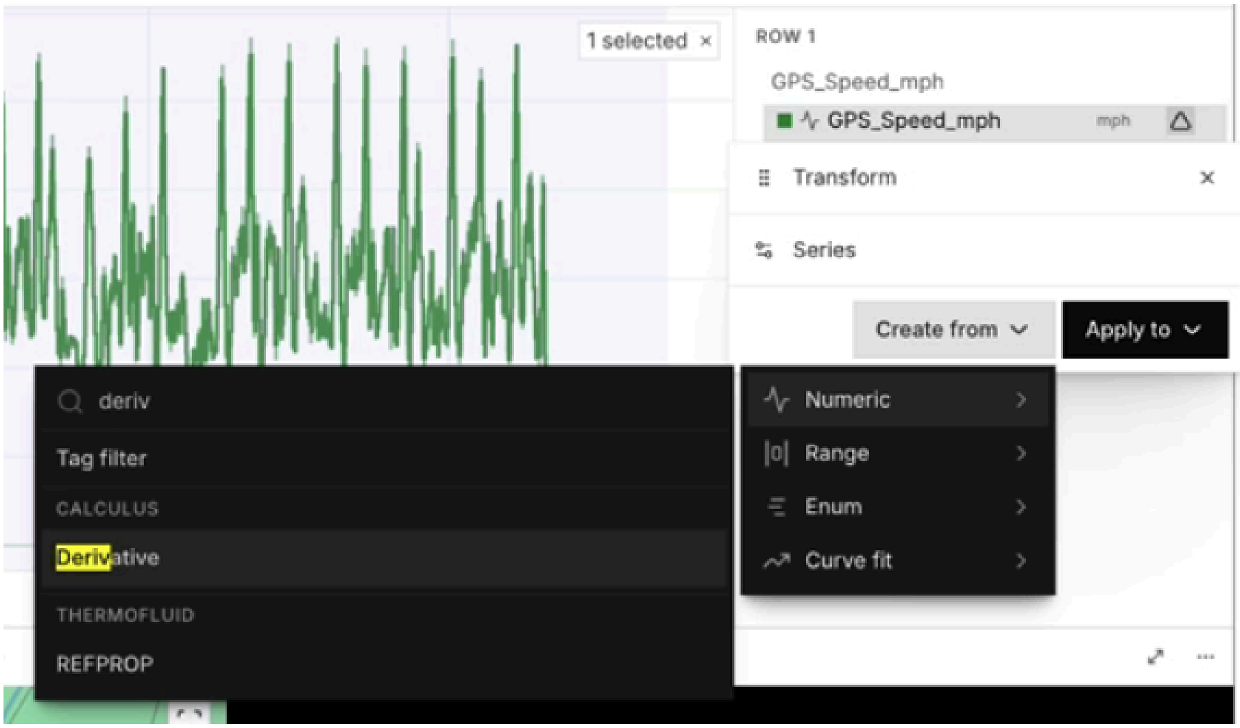
1. In the workbook, select the channel you would like to transform and load it into the time series panel.



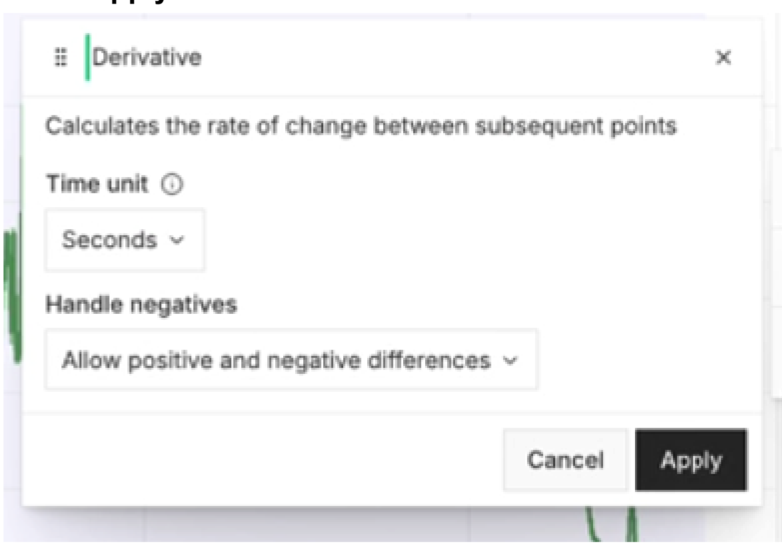
2. Select the Δ icon next to the name of the channel.



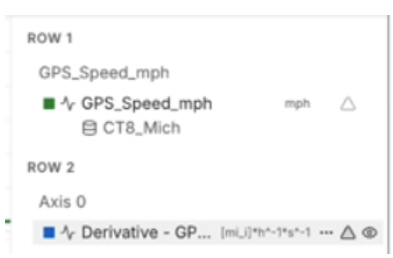
3. Under the **Transform** panel, select the **Create from** drop-down. We are creating a new derived variable so we select **Create from** instead of **Apply to**. Select **Numeric** and type in “derivative” in the search bar to select it.



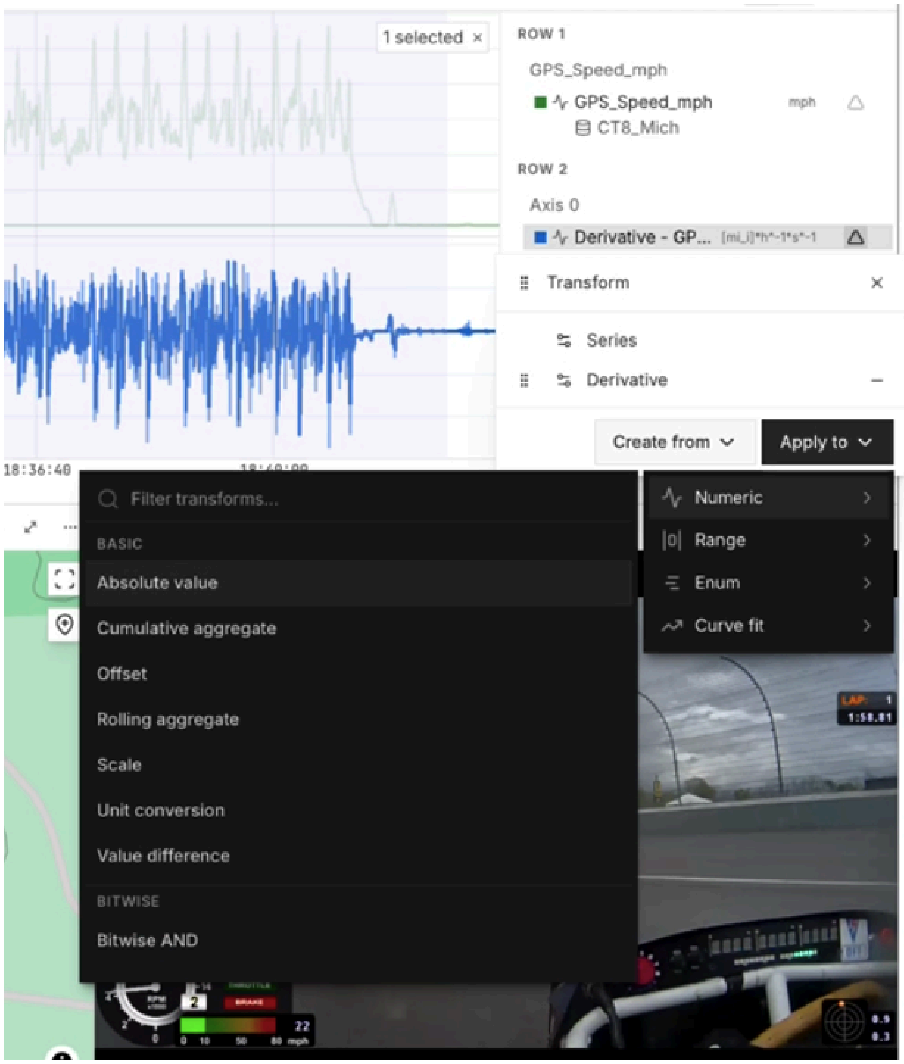
4. Select **Apply.**



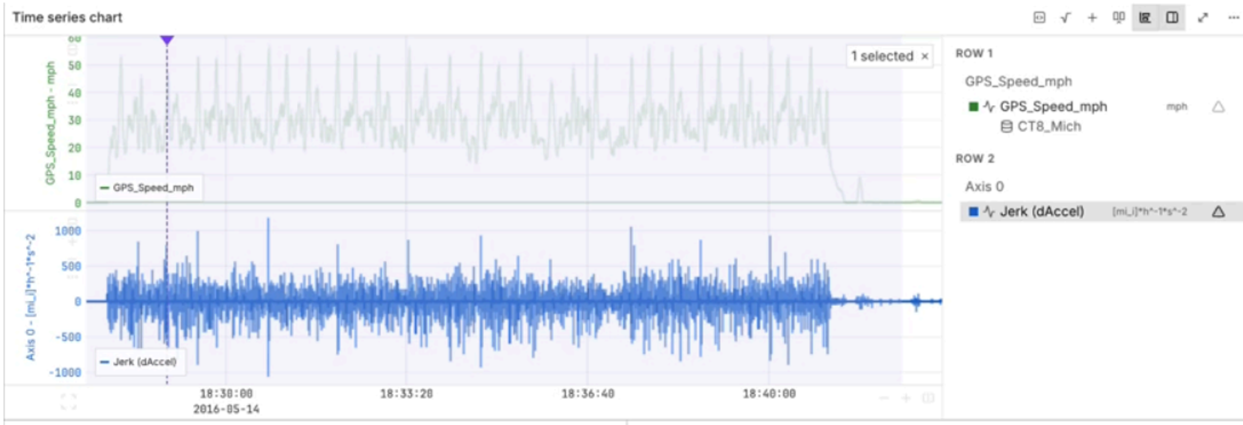
This creates a new row in our series panel with our new variable (the derivative we just calculated). Its default name is "Derivative - [Name of Channel]" but you can change it if you'd like.



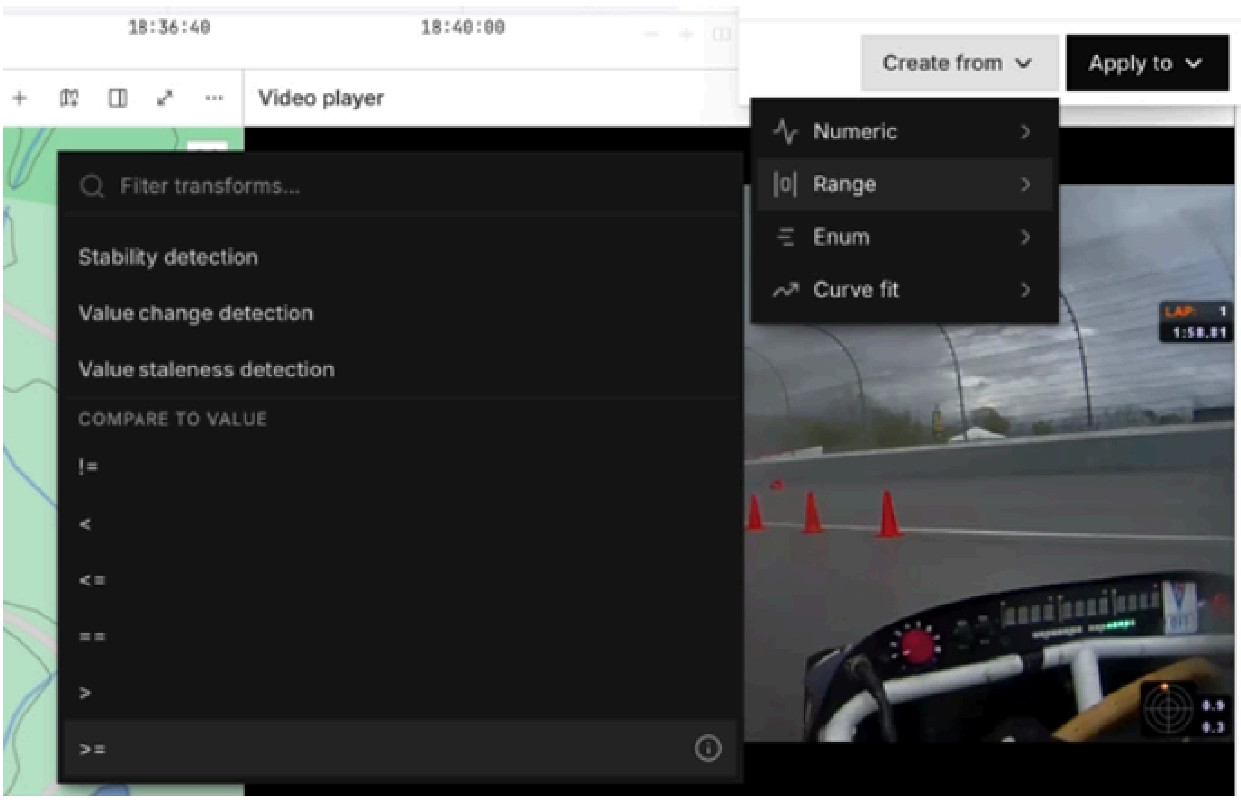
5. We now have transformed our speed data into the derivative of speed (acceleration). To calculate jerk, we will transform our derivative. Select ∆ next to the channel name. Since we are transforming on top of an existing variable this time, we will use **Apply to** instead of **Create from**. This will alter the existing graph in the series panel.



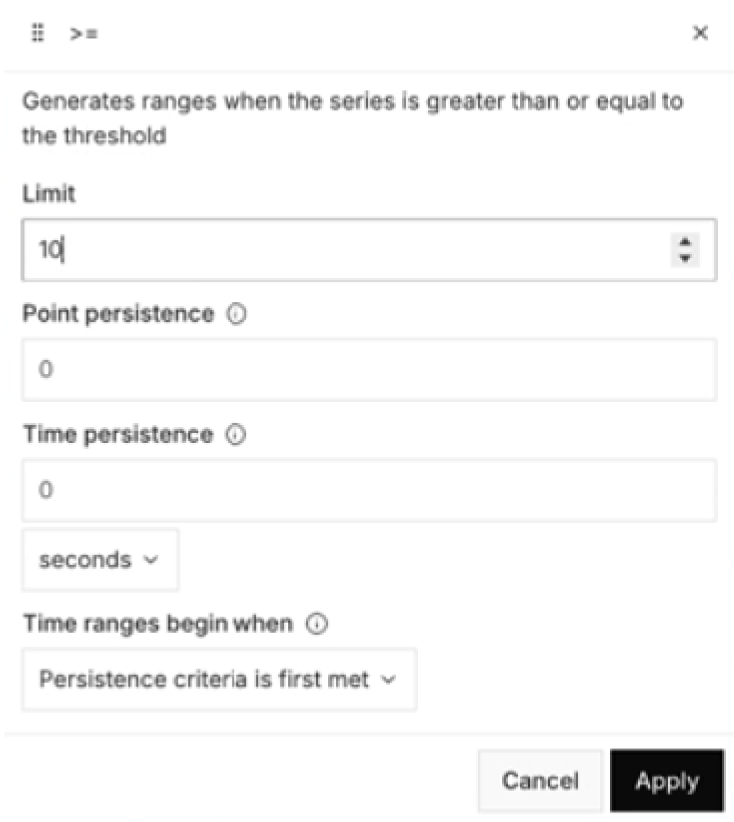
6. Select **Numeric** and type "derivative" once more into the search bar to select. Hit **Apply** when prompted.
7. We have now created the derivative of the derivative of our original speed data using **Transform** functionality. Let's rename it "Jerk(dAccel)" for clarity.



8. We will now create a range for our jerk data greater than or equal to 10. Repeat step 5. Now, instead of **Numeric** select **Range**. Select **>=** from the transforms list.



9. In the modal set the **Limit** equal to 10 and hit **Apply.**



We have now shown how to transform a time series numerically and add a range.

A Short Note

I chose a guided tutorial with screenshots to show off the functionality of transform. I chose not to include the part of the video with the formula using the left and right wheel because I found it to be distracting to the reader. Instead, I took the relevant part of that section (showcasing range) and included just that in my tutorial. Before publishing, I'd like to see the other transform functions, specifically Enum and Curve fit.

AI Usage

I think it's important to *strategically* use AI, and for this assignment I did not feel it could enhance the final product. Therefore, I did not use AI.